NRSG 233 HEALTH ASSESSMENT · MODULE 4

Abdomen & Musculoskeletal System Complete study guide

For Luke's Quiz 4 (Module 4 content) on Tuesday, October 20, 2026, 8:30 am PT.

> **Personal study sheet, not for submission.** NRSG 233 bars AI on submitted work. The Module 4 documentation sheet is not filled in here.

Built from: (1) Dr. Sabine Dunbar's lecture slides, "Module 4 Abdomen & Musculoskeletal System" (F26), as the **primary source**, in slide order, cited [Slide N]; (2) her assigned Jarvis & Eckhardt 9th ed. readings: Ch. 22 Abdomen, Objective Data (pp. 542–561) and pp. 563–565; Ch. 23 Musculoskeletal, Objective Data (pp. 585–613); (3) her Module 4 vocabulary list (18 terms; definitions written from her slides and Jarvis).

Blue boxes = her slides. ⚠ boxes = places where her slides and Jarvis differ; hers come first. Her handout PDF skips slides 2–5, so numbering jumps from 1 to 6.

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★ What she emphasized

Her objectives — the "you will be able to" list [Slide 6]

- Identify underlying abdominal organs by landmarks and the **four quadrants** [Slide 6].
- Perform abdominal assessment in the order **inspection, auscultation, percussion, palpation** [Slide 6].
- Demonstrate **inspection and palpation of joints bilaterally** [Slide 6].
- Assess **muscle strength and active range of motion** [Slide 6].
- Distinguish normal vs abnormal findings and **use appropriate terminology to document** [Slide 6].
- Describe **appropriate nursing actions for any abnormal findings** [Slide 6].

Preparation items she listed [Slide 7]

- Good lighting; relaxed, comfortable client; **client empties bladder** [Slide 7].
- Supine, arms at sides or across chest, knees bent or on pillow; drape appropriately [Slide 7].
- **Ask about tenderness first**; warm hands and stethoscope; short fingernails [Slide 7].

Her documentation-sheet items (the things you actually chart)

- **Contour** — flat, rounded, scaphoid, protuberant; profile from rib margin to pubic bone; normally flat to rounded [Slide 11; doc sheet, Inspection p. 542].
- **Symmetry** — symmetric bilaterally; note bulging, mass, asymmetry; deep breath may reveal masses/enlarged organs [Slide 12].
- **Umbilicus** — midline, inverted, no discoloration, inflammation, or hernia [Slide 12].
- **Skin** — smooth, even, homogeneous color; nevi common; scars: document cause, location, length; fine venous network in thin persons; striae = one common pigment change [Slide 13].
- **Pulsations/movement** — aortic pulsation may be visible in epigastric area in thin persons; respiratory movement; peristalsis visible in very thin persons [Slide 15].
- **Bowel sounds: Present / Hyperactive / Hypoactive** — auscultate with diaphragm, character & frequency, **begin in RLQ**; normal = high-pitched, gurgling, cascading; hyperactive = loud, high-pitched, rushing, tinkling (increased motility); hypoactive/absent = surgery, peritonitis, paralytic ileus, complete obstruction; absence established only after **5 minutes** of continuous auscultation [Slide 17; doc sheet p. 545].
- **Bruits** — auscultate aorta, renal, iliac, femoral arteries [Slide 19].
- **Percussion — overall sound** (tympany / dullness / hyperresonance) [Slide 20; doc sheet p. 546].
- **Light palpation findings** — depress ~1 cm; overall impression of skin surface and superficial musculature; note resistance, **muscle guarding, tenderness**; normally soft & smooth [Slide 23; doc sheet p. 548].
- Deep palpation ~**5–8 cm**, bimanual for obese/large abdomen; note location, size, consistency, mobility, enlargement, tenderness, masses [Slides 24–25].
- **Rebound tenderness** — press slowly and deeply at 90° in a non-tender area, withdraw quickly; normally no pain =

peritoneal irritation/inflammation if pain [Slide 26].

- Abnormal findings to recognize: abdominal distention, abdominal pain, hernia, abnormal bowel sounds, enlarged organs [Slide 27].

MSK items she emphasized

- Assessment of **joints, muscles, bones** plus **functional assessment of ADLs** [Slide 28].
- Joint types: fibrous (no movement, skull sutures), cartilaginous (slightly moveable, vertebrae), synovial (freely moveable, synovial membrane secretes fluid) [Slide 29].
- **Joint inspection** — size & contour; skin/tissue color, swelling, masses, deformity [Slide 30; doc sheet p. 585].
- **Joint palpation** — skin temperature, muscles, bony articulations, joint capsule; joints normally non-tender [Slide 31; doc sheet p. 586].
- **ROM: Full / Active / Limited** — do **active ROM first**; if limitation, do passive ROM; compare bilaterally; normally no tenderness, pain, or crepitation [Slide 32; doc sheet].
- Movement terms she defined: flexion, extension, abduction, adduction, pronation, supination, circumduction, inversion, eversion, rotation [Slide 33].
- ROM head-to-toe order: TMJ, cervical spine, shoulder, elbow, wrist & hand, hip, knee, ankle & foot, spine [Slide 35].
- **Muscle strength** — test through same motions as AROM; equal bilaterally, fully resists opposing force; **grade 0–5** per grading scale p. 587 [Slide 37; doc sheet].
- Documentation grid: ROM rated Right/Left for each joint group; muscle strength numeric 0–5 for deltoids, biceps, triceps, wrist, fingers, hips, quads, hamstrings, ankles/feet [doc sheet].

⚠ **Conflict:** Her slide says grade muscle strength on a "**5-point scale**" [Slide 37]; the grading scale is actually **0–5 (six grades)**, and her own documentation sheet says "numeric value from 0–

5 ... Muscle Grading Scale p. 587" [doc sheet]. Use the 0–5 scale as printed in Jarvis [Jarvis p. 587].

⚠ **Conflict:** She instructs auscultating in all four quadrants and establishing absence after 5 minutes of continuous auscultation to 4 quadrants [Slide 17]. Jarvis says sounds radiate widely, so listening in all four quadrants "is not necessary" — just judge present / hypoactive / hyperactive — but agrees on the 5-minute rule for absence [Jarvis p. 545]. Her rule is stricter; follow her on the exam. Jarvis also notes early research suggests auscultating first may not be essential [Jarvis p. 545 footnote], but her exam order remains inspection → auscultation → percussion → palpation [Slide 6; Slide 17].

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0. Her objectives

Module 4: Abdomen & Musculoskeletal System — LLUSN NRSG 233, Sabine Dunbar, DNP, NMW, RN, CNE [Slide 1]

Objectives [Slide 6]: At the end of this module you will be able to:

1. Identify underlying abdominal organs in relation to landmarks of the abdomen and the four quadrants.
2. Perform abdominal assessment through inspection, auscultation, percussion, and palpation.
3. Demonstrate inspection and palpation of joints bilaterally.
4. Assess muscle strength and active range of motion.
5. Distinguish normal vs. abnormal findings of the abdomen and musculoskeletal exam; use appropriate terminology to document.
6. Describe appropriate nursing actions for any abnormal findings.

Note: Slides 2–5 were not provided; slides 1, 6–27 are the abdomen content reproduced below. Musculoskeletal objectives (3–4) are covered elsewhere in the module; Jarvis assigned pages here cover only the abdomen [Jarvis pp. 542–565].

1. Abdomen: preparation

Preparation [Slide 7]:

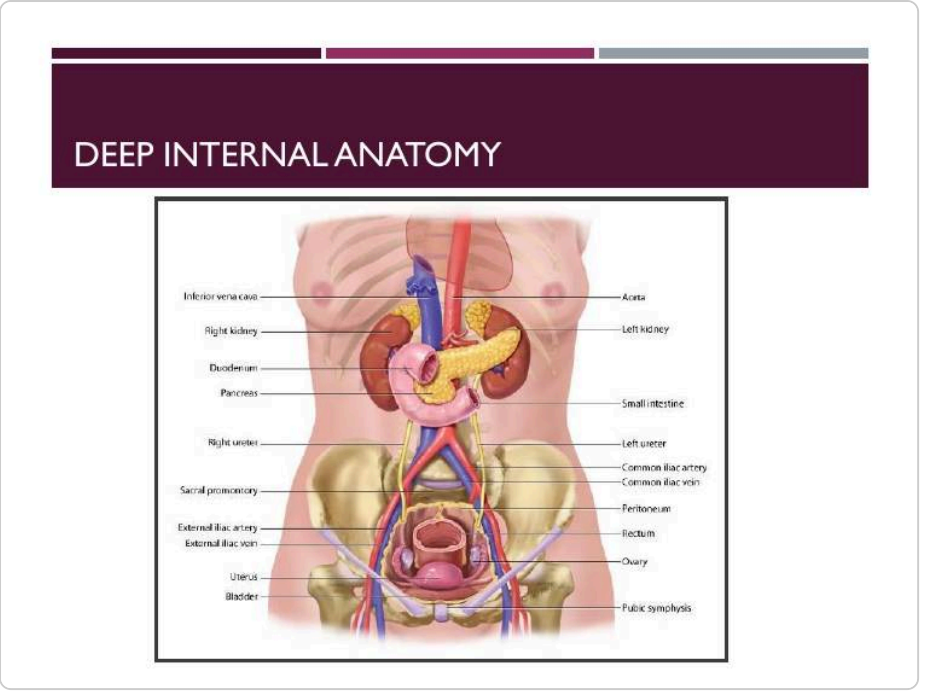
- Good lighting
- Relaxed, comfortable client
- Have client empty bladder
- Supine, arms at sides or across chest, knees bent or on pillow
- Drape client appropriately
- Ask regarding presence of tenderness
- Warm hands & stethoscope; short fingernails

Jarvis fill-in [Jarvis p. 542]:

- Lighting: strong overhead light *plus* a secondary stand light. Expose abdomen fully; drape genitalia and female breasts.
- Bladder emptied, saving a urine specimen if needed; keep room warm to avoid chilling/tensing.
- Head on a pillow; **discourage arms over the head** — this tenses abdominal musculature.
- Examine painful areas **last** to avoid muscle guarding.
- **Distraction techniques** for relaxation: slow breathing, emotive imagery, low soothing voice, conversation, or having the person relate abdominal history while you palpate [Jarvis p. 548].
- Equipment: stethoscope, alcohol wipe to clean endpiece [Jarvis p. 542].

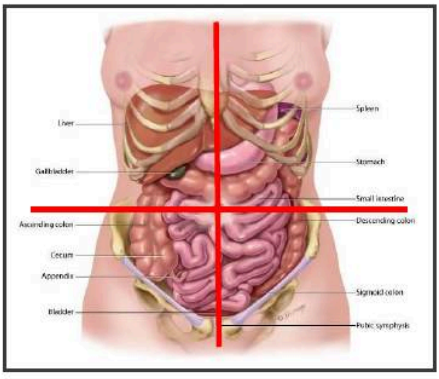
2. Anatomy: four quadrants and nine regions

Anatomy [Slide 8]: diagram of **Four Quadrants** and **Nine Regions** (slide is a picture; reference cited on slide is Seidel/Mosby, 6th & 7th eds.).
Slide 9: picture of organs by quadrant — labeled "ORGANS/QUADRANT" (no text).
Slide 10: picture of deep internal anatomy (no text).



Deep internal anatomy [Slide 10]

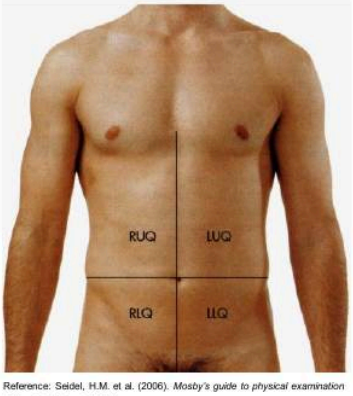
ORGANS/QUADRANT



Organs by quadrant [Slide 9]

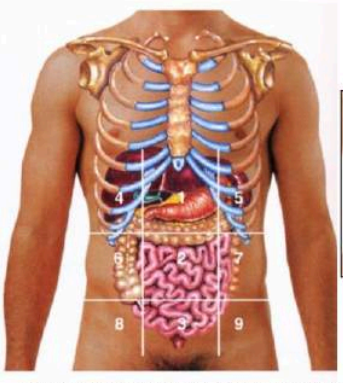
ANATOMY

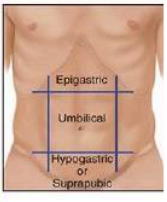
Four Quadrants



Reference: Seidel, H.M. et al. (2006). *Mosby's guide to physical examination* (6th ed.). St. Louis: Mosby

Nine Regions





Reference: Seidel, H.M. et al. (2011). *Mosby's guide to physical examination* (7th ed.). St. Louis: Mosby.

Four quadrants and nine regions [Slide 8]

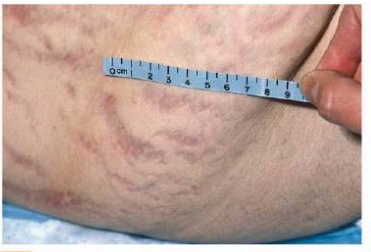
Jarvis fill-in — structures named in the assigned pages [Jarvis pp. 545, 550–553]:

Quadrant / area	Structures named in Jarvis
RUQ	Liver (edge, lower border), gallbladder (under liver border), right kidney lower pole, ascending colon
RLQ	Cecum, ileocecal valve area (bowel sounds always present here), appendix region (McBurney point)
LUQ	Spleen (not normally palpable), left kidney (≈1 cm higher than right, not normally palpable)
LLQ	Sigmoid colon (mild tenderness normally present)

Other landmarks/structures [Jarvis Fig. 22.18, p. 550]: xiphoid process; rectus muscles (lateral borders); sacral promontory; pulsatile aorta (upper abdomen, slightly left of midline); full bladder and gravid uterus (lower midline). Normally palpable structures are listed in Fig. 22.18 [Jarvis p. 550].

⚠ **Note:** Her slides show the quadrant/region diagrams as pictures only [Slides 8–10] and cite Seidel/Mosby. The table above is filled only with structures explicitly named in the assigned Jarvis pages — verify the full quadrant organ list against your lecture diagram/Seidel.

3. Inspection

Normal Range of Findings	Abnormal Findings
Symmetry Shine a light across the abdomen toward you or lengthwise across the person. The abdomen should be symmetric bilaterally (Fig. 22.8). Note any localized bulging, visible mass, or asymmetric shape. Even small bulges are highlighted by shadow. Step to the foot of the examination table to recheck symmetry.	Bulges, masses. Hernia—Protrusion of abdominal viscera through abnormal opening in muscle wall (see Table 22.4, Abnormalities on Inspection, p. 567). Sister Mary Joseph nodule is a hard nodule in umbilicus that occurs with metastatic cancer of stomach, large intestine, ovary, or pancreas.[15]
 22.8 Ask the person to take a deep breath to further highlight any change. The abdomen should stay smooth and symmetric. Or ask the person to perform a sit-up without pushing up with their hands.	Note any localized bulging. Hernia or enlarged liver or spleen may show.
Umbilicus Normally it is midline and inverted, with no sign of discoloration, inflammation, or hernia. It becomes everted and pushed upward with pregnancy. The umbilicus is a common site for piercings. The site should not be red or crusted.	Everted with ascites or underlying mass (Table 22.1). Deeply sunken with obesity. Enlarged, everted with umbilical hernia. Bluish periumbilical color occurs (though rarely) with intraperitoneal bleeding (Cullen sign).[15]
Skin The surface is smooth and even, with homogeneous color. This is a good area to judge pigment because it is often protected from sun.	Redness with localized inflammation. Jaundice (shows best in natural daylight). Skin glistening and taut with ascites. Striae also occur with ascites.
One common pigment change is **striae** (lineae albicantes)—silvery white, linear, jagged marks about 1 to 6 cm long (Fig. 22.9). They occur when elastic fibers in the reticular layer of the skin are broken after rapid or prolonged stretching as in pregnancy or excessive weight gain. Recent striae are pink or blue; then they turn silvery white.  22.9 Striae.	Striae look purple-blue with Cushing syndrome (excess adrenocortical hormone causes the skin to be fragile and easily broken from normal stretching).

Objective Data

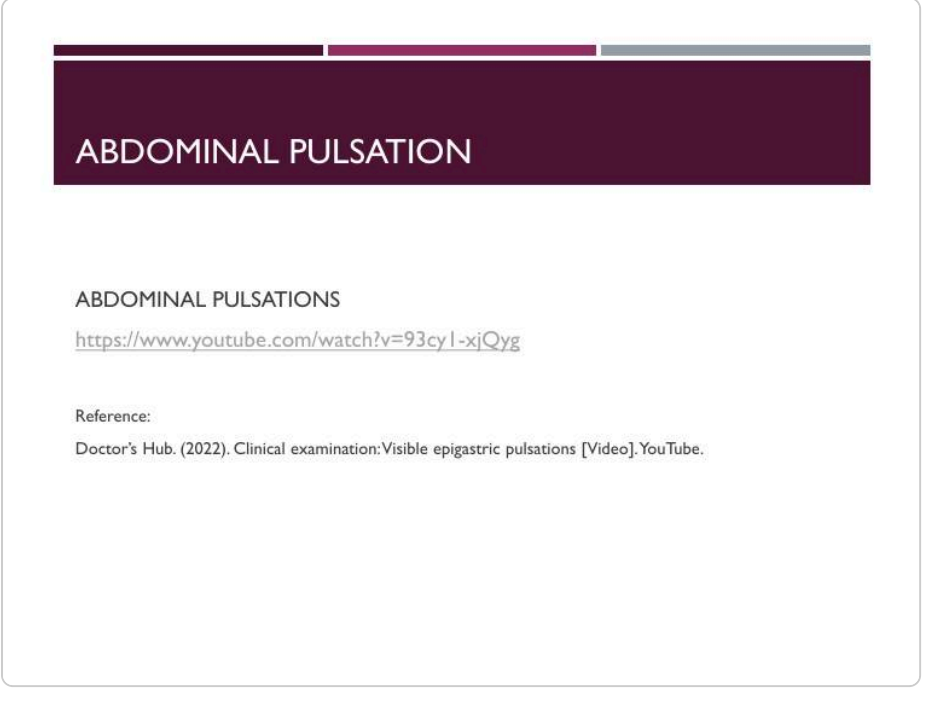
Fig. 22.8–22.9: localized bulging; striae [Jarvis p. 543]

Contour [Slide 11]: Determine profile from rib margin to pubic bone. Contour describes nutritional state and normally ranges from **flat to rounded**.

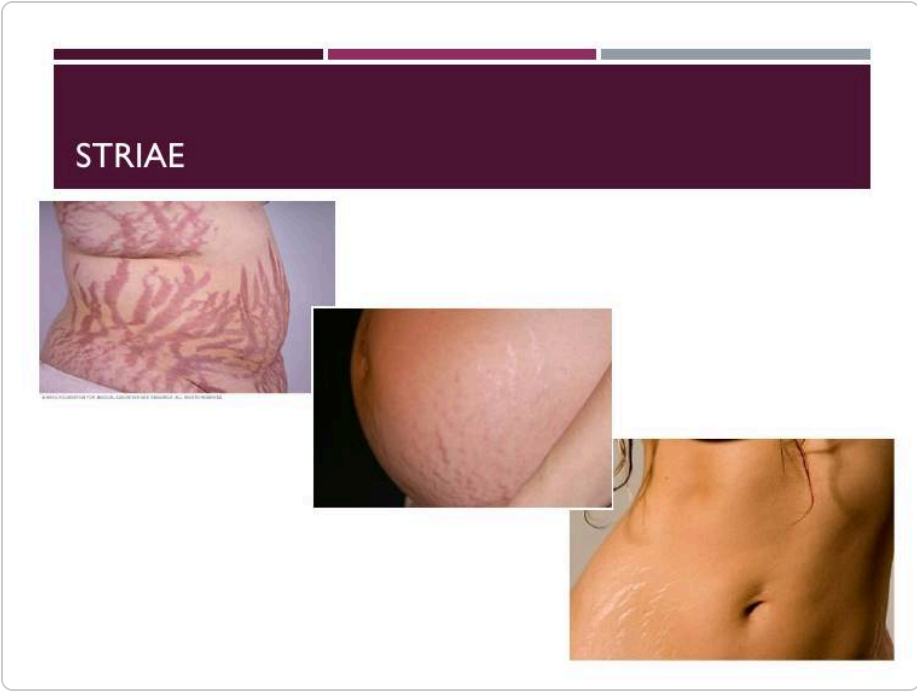
Symmetry [Slide 12]: abdomen should be symmetric bilaterally; note localized bulging, visible mass, or asymmetric shape; ask client to take a deep breath — may make masses and enlarged organs more visible.
Umbilicus [Slide 12]: normally midline and inverted, no discoloration, inflammation, or hernia.

Skin [Slide 13]: surface smooth and even, homogeneous color; pigmented nevi (moles) — circumscribed brown macular or papular areas — common on abdomen; lesions/rashes normally absent; surgical scars may be present — assess cause, document location and length; fine venous network may be visible in thin persons; one common pigment change is **striae**.

Slide 14: STRIAE — picture only (photo of striae).
Slide 16: Abdominal pulsation — video link (Doctor's Hub, 2022, visible epigastric pulsations).



Abdominal pulsation [Slide 16]



Striae [Slide 14]

Pulsation or movement [Slide 15]: normally you may see aortic pulsations beneath skin in the **epigastric** area, particularly in thin persons with good muscle wall relaxation; respiratory movement also shows in abdomen, particularly in males; waves of peristalsis sometimes visible in very thin persons.

Jarvis fill-in [Jarvis pp. 542–544]:

- **Contour:** stand on person's right side, look down, then stoop/sit to gaze across (head slightly higher than abdomen). **Scaphoid** = abdomen caves in; **protuberant** = distention (Table 22.1, p. 563). Contour options for charting: flat / rounded / scaphoid / protuberant.
- **Symmetry:** shine a light across the abdomen toward you — shadows highlight even small bulges; recheck from foot of table. Ask for deep breath *or* a sit-up without pushing with hands. Hernia = protrusion of abdominal viscera through abnormal opening in muscle wall. **Sister Mary Joseph nodule** = hard umbilical nodule with metastatic cancer of stomach, large intestine, ovary, or pancreas.
- **Umbilicus:** everted with ascites or underlying mass; deeply sunken with obesity; enlarged/everted with umbilical hernia; bluish periumbilical color (rare) = **Cullen sign** (intraperitoneal bleeding); everted and pushed upward with pregnancy. Piercing site should not be red or crusted.
- **Skin:** good area to judge pigment (sun-protected). **Striae (lineae albicantes)**: silvery-white, linear, jagged, 1–6 cm long, from broken elastic fibers after rapid/prolonged stretching (pregnancy, excessive weight gain); recent = pink or blue → silvery white; purple-blue with Cushing syndrome; also occur with ascites. Skin glistening and taut with ascites; redness = localized inflammation; jaundice (best in natural daylight).
- Scars: draw location in record, indicate length in cm; scars alert you to possible underlying adhesions and excess fibrous tissue.
- Veins: fine venous network normal in thin persons; prominent dilated veins (**caput medusae**) = portal hypertension, cirrhosis, ascites, or vena caval obstruction; veins more visible with malnutrition.
- **Turgor:** pinch a skin fold, release — poor turgor with dehydration (common with GI disease) [Jarvis p. 544].
- **Pulsation/movement:** as her slide; marked aortic pulsation with widened pulse pressure (hypertension, aortic insufficiency, thyrotoxicosis) and aortic aneurysm; marked visible peristalsis *with* distention = intestinal obstruction.

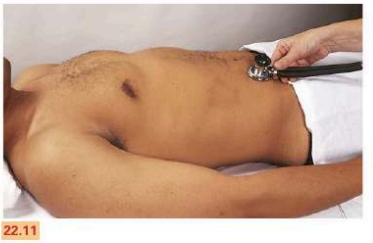
- **Also inspect:** hair distribution (diamond pattern adult males, inverted triangle adult females) [Jarvis p. 544].
- **Demeanor:** normal = relaxed, benign expression, slow even respirations. Restlessness/constant turning = colicky pain (gastroenteritis, bowel obstruction); **absolute stillness resisting movement = pain of peritonitis**; knees flexed, facial grimacing, rapid uneven respirations indicate pain [Jarvis pp. 544–545].

4. Auscultation

Normal Range of Findings	Abnormal Findings
	Absolute stillness, resisting any movement, occurs with the pain of peritonitis. Knees flexed up, facial grimacing, and rapid, uneven respirations also indicate pain.

Auscultate Bowel Sounds and Vascular Sounds

Depart from the usual examination sequence and auscultate the abdomen next. This is done because percussion and palpation can increase peristalsis, which would give a false interpretation of bowel sounds.[a] Use the diaphragm endpiece because bowel sounds are relatively high-pitched. Hold the stethoscope lightly against the skin; pushing too hard may stimulate more bowel sounds (Fig. 22.11). Begin in the RLQ at the ileocecal valve area because bowel sounds normally are always present here.



22.11

Bowel Sounds

Note the character and frequency of bowel sounds. Although the origin of bowel sounds is not fully understood, they may originate from the movement of air and fluid within the stomach and large and small intestine. A wide range of normal sounds can occur. Normal bowel sounds are high-pitched, gurgling, cascading sounds, occurring irregularly anywhere from 5 to 30 times per minute. Do not bother to count them. In addition, because the sounds radiate widely over the abdomen, the gurgle you hear in the RLQ may originate in the stomach. Therefore listening in all four quadrants is not necessary.[6,15] Just judge if they are present or are hypoactive or hyperactive.

One type of hyperactive bowel sounds is fairly common: hyperperistalsis, when you feel your "stomach growling," termed **borborygmus.**

Bowel sound interpretation is highly subjective and can vary widely among clinicians, and bowel sounds are likely not a reliable indicator of bowel function in many circumstances.[6,7] For example, should you withhold oral intake following following colorectal surgery? Other indicators may be more reliable (tolerance of oral intake, passage of stool, and lack of abdominal distension).[3,10] A perfectly "silent abdomen" is uncommon; you must listen for 5 minutes by your watch before deciding whether bowel sounds are completely absent.

Two distinct patterns of abnormal bowel sounds may occur:

1. **Hyperactive sounds** are loud, high-pitched, rushing, tinkling sounds that signal increased motility.
2. **Hypoactive or absent sounds** follow abdominal surgery or with inflammation of the peritoneum (see Table 22.5, Abnormal Bowel Sounds, p. 568, and Table 22.2, p. 565).

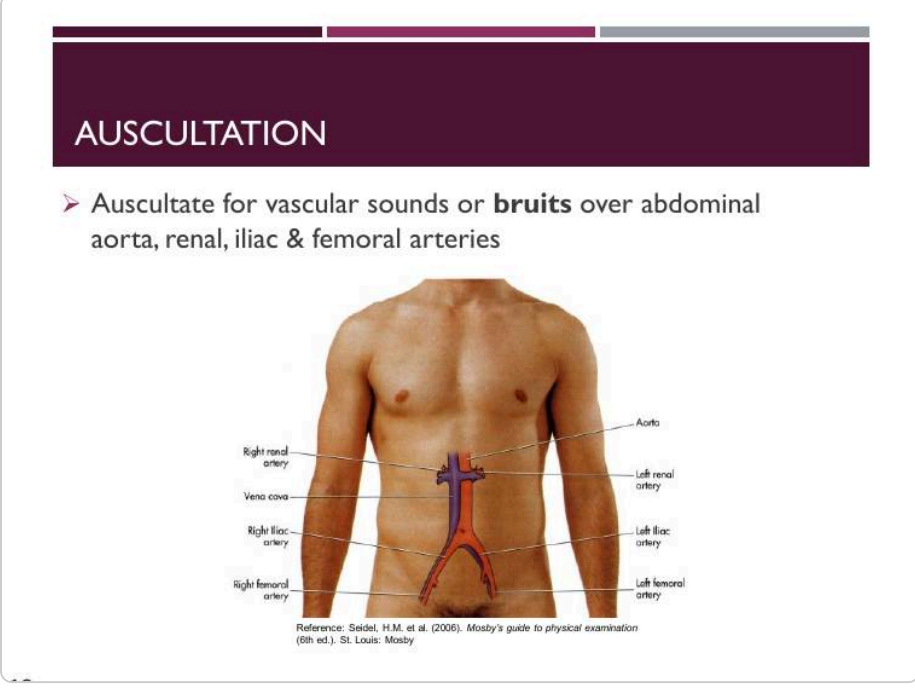
[a]Early research suggests this change in the order of examination sequence may not be necessary. That is, abdominal palpation did not change the number of bowel sounds in a small number of healthy people and in GI outpatients. Further research is necessary.[26]

Fig. 22.11: listening for bowel sounds [Jarvis p. 545]

Auscultation [Slide 17]:

- Auscultate with **diaphragm**, noting character & frequency of bowel sounds, beginning in **RLQ**.
- Normal sounds: high pitched, gurgling, cascading.
- **Hyperactive BS** — loud, high pitched, rushing, tinkling, d/t increased motility: early obstruction, gastroenteritis, diarrhea, laxatives, subsiding paralytic ileus.
- **Hypoactive or absent BS** — d/t abdominal surgery, peritonitis, paralytic ileus, or complete obstruction.
- Establish absence only after **5 minutes** of continuous auscultation to 4 quadrants.

Slide 18: BOWEL SOUNDS — video link (bowel sounds clip).
Vascular sounds/bruits [Slide 19]: auscultate over abdominal aorta, renal, iliac & femoral arteries (slide cites Seidel/Mosby).



Vascular sounds: where to listen for bruits [Slide 19]

Jarvis fill-in [Jarvis pp. 545–546]:

- Depart from the usual sequence and auscultate **next after inspection** — percussion and palpation can increase peristalsis and falsely alter bowel sounds (early research suggests this may not be necessary; further research needed).
- Use diaphragm (sounds are high-pitched); hold lightly — pressing hard may stimulate more sounds. Begin in RLQ at the **ileocecal valve area** — bowel sounds are always present here.
- Normal: high-pitched, gurgling, cascading, irregularly **5–30 times per minute**; do not count them. Sounds radiate widely, so listening in all four quadrants is **not necessary** — judge present / hypoactive / hyperactive.
- **Borborygmus** = hyperperistalsis ("stomach growling") — a fairly common type of hyperactive sound.
- Interpretation is highly subjective and may not reliably indicate bowel function (e.g., after colorectal surgery, tolerance of oral intake, passage of stool, and lack of distention may be more reliable). A truly "silent abdomen" is uncommon — listen **5 minutes by your watch** before declaring sounds absent.
- **Vascular sounds:** firmer pressure over aorta, renal, iliac, femoral arteries — especially with hypertension. Usually no sound; **4%–20% of healthy people** (usually <40 y) may have a normal bruit, usually from the celiac artery — systolic, medium-to-low pitch, heard between xiphoid and umbilicus. Note location, pitch, timing. Systolic bruit = pulsatile blowing sound with stenosis, partial occlusion, or aneurysm. Venous hum and peritoneal friction rub are rare.
- **Safe practice:** do NOT use auscultation of an air bolus to verify nasogastric tube placement — confirm by chest x-ray; ongoing checks by external tube length measurement and aspirate pH <5.5.

⚠ **Conflict:** Her slide says to auscultate "to 4 quadrants" [Slide 17]; Jarvis states listening in all four quadrants is not necessary because sounds radiate widely [Jarvis p. 545]. **Hers first** for lab

check-off (auscultate all four quadrants, starting RLQ). Both agree on the 5-minute rule before declaring sounds absent.

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5. Percussion

Percussion [Slide 20]: assesses relative density of abdominal contents, locates organs, screens for abnormal fluid or masses.

- **Tympany** — high pitched, musical; presence of air in stomach & intestine.
- **Dullness** — medium intensity, duration & pitch; solid organs, masses, fluid, adipose tissue, distended bladder.
- **Hyperresonance** — very loud, low pitch, long duration; gaseous distention.

Not recommended for determining liver span, splenomegaly, or bladder distention [Slide 21].
Costovertebral angle (CVA) percussion for kidney tenderness [Slide 21].

Jarvis fill-in [Jarvis pp. 546–547, 554–555]:

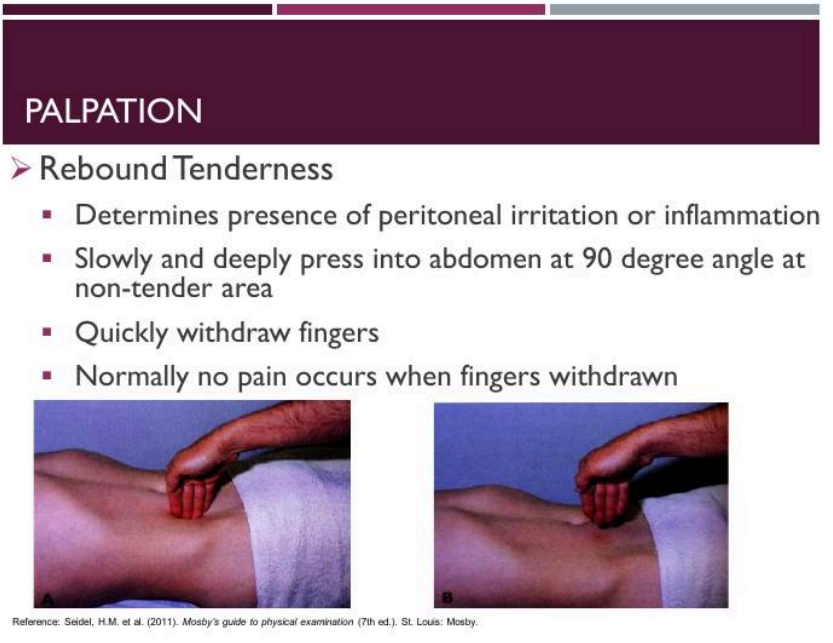
- Percuss lightly in all four quadrants, moving **clockwise**, to determine prevailing tympany vs. dullness. **Tympany should predominate** — air rises to the surface when supine.
- Dullness over a distended bladder, adipose tissue, fluid, or a mass; hyperresonance with gaseous distention.
- **Liver span by percussion: not recommended** — underestimates true liver size (upper border placed too low and/or lower border too high), variable between examiners, often misses hepatomegaly. Assess liver by **palpation** [Jarvis p. 550].
- **Splenic dullness percussion omitted** — palpation is more reliable. **Bladder percussion omitted** as unreliable — use bedside bladder scanning ultrasound.
- **CVA tenderness:** place one hand over the 12th rib at the CVA on the back; thump it with the ulnar edge of the other fist (indirect fist percussion). Normal = a thud, no pain. **Sharp pain** = inflammation of kidney or perinephric area (e.g., pyelonephritis). Usually done with thoracic assessment, person seated, examiner behind.
- **Ascites percussion tests (advanced practice)** [Jarvis pp. 554–555]: **fluid wave** — firm hand in midline blocks skin transmission, strike opposite flank; positive = distinct tap felt (large ascites). **Shifting dullness** — percuss supine from tympany over top down to dullness at the flank fluid level, mark it, roll person to right side and repeat; level of dullness shifts upward. Positive only with large volumes — will not detect <**500–1100 mL**. Ultrasound is the definitive tool.

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6. Palpation

Purpose [Slide 22]: assess size, location, consistency of certain organs; screen for abnormal mass or tenderness. **Assess tender areas last.** Observe non-verbal cues of discomfort.

Light palpation [Slide 23]: depress abdominal wall **~1 cm**; objective is not to search for organs but to form an overall impression of skin surface and superficial musculature; note increased resistance or muscle guarding and tenderness (normally abdomen is soft & smooth).
Deep palpation [Slide 24]: depress **~5–8 cm**; use **bimanual technique** for very large or obese patient.
Deep palpation [Slide 25]: delineate organs & masses; note location, size, consistency, and mobility of palpable organs and presence of abnormal enlargement, tenderness, or masses.
Rebound tenderness [Slide 26]: determines peritoneal irritation/inflammation; slowly and deeply press into abdomen at **90° angle** at a **non-tender area**; quickly withdraw fingers; normally no pain on withdrawal.



Rebound tenderness [Slide 26]

PALPATION

- Deep Palpation
  - Depress abdominal wall ~ 5-8 cm
  - Use bimanual technique for very large or obese patient



FIGURE 17-13
Deep palpation of the abdomen. Press deeply and evenly with the palmar surface of extended fingers.

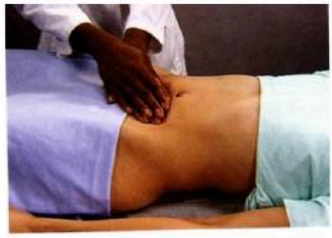


FIGURE 17-14
Deep bimanual palpation.

Reference: Seidel, H.M. et al. (2011). *Mosby's guide to physical examination* (7th ed.). St. Louis: Mosby.

Deep palpation [Slide 24]

PALPATION

- Light Palpation
  - Depress abdominal wall ~ 1cm
  - Objective is not to search for organs but to form an overall impression of skin surface and superficial musculature
  - Note increased resistance or muscle guarding & tenderness (**normally abdomen is soft & smooth**)



Reference: Seidel, H.M. et al. (2011). *Mosby's guide to physical examination* (7th ed.). St. Louis: Mosby.

Light palpation [Slide 23]

Jarvis fill-in [Jarvis pp. 548–553, 555–556]:

- **Relaxation measures** before palpating: bend knees; keep palpating hand **low and parallel** to abdomen; slow breathing (in nose, out mouth); low soothing voice; emotive imagery; for ticklish people, place their hand under yours with your fingers curled over theirs (people are not ticklish to themselves), or palpate around/with the stethoscope right after auscultation.
- **Light palpation:** first four fingers close together, depress ~1 cm, gentle rotary motion sliding fingers and skin together, **lift** (don't drag), move clockwise. Save tender areas for last so pain/rigidity doesn't obscure deep palpation.
- **Voluntary guarding** (cold, tense, ticklish): bilateral, relaxes slightly on exhalation — use relaxation measures. **Involuntary rigidity**: constant, boardlike hardness; may be unilateral; protective with acute peritoneal inflammation; the same area hurts when the person attempts a sit-up.
- **Deep palpation:** push down 5–8 cm (2–3 in), clockwise, entire abdomen. **Bimanual**: hands stacked — top hand pushes, bottom hand palpates. Note location, size, consistency, mobility of organs; abnormal enlargement, tenderness, masses. Some structures are normally palpable (Fig. 22.18).
- Mild tenderness over the **sigmoid colon (LLQ)** is normal; any other tenderness is investigated. If a mass is found: describe **location, size, shape, consistency (soft/firm/hard), surface (smooth/nodular), mobility (incl. with respirations), pulsatility, tenderness** [Jarvis p. 550].
- **Liver** [Jarvis pp. 550–551]: left hand under the back at 11th–12th ribs lifts/supports; right hand on RUQ, fingers parallel to midline, push deeply under right costal margin; with each exhalation move up 1–2 cm; normal = firm regular ridge bumping fingertips on inhalation, or not palpable at all. Liver >**1–2 cm** below right costal margin = enlarged (except with depressed diaphragm); record cm, consistency, tenderness. Variations: chronic emphysema displaces liver downward but size normal. Alternatives: **hooking technique** (hook fingers over costal margin from above, deep breath); **scratch test** (stethoscope at xiphoid, stroke finger up MCL from RLQ

— sound magnified at liver edge; useful when abdomen is distended, obese, too tender, or guarded).

- **Spleen** [Jarvis p. 552]: normally **not palpable** — must be enlarged **3× normal size** to be felt. Left hand behind left side at 11th–12th ribs lifts; right hand oblique on LUQ, fingers toward left axilla, push under left costal margin, deep breath. Alternative: roll person onto **right side**. If enlarged spleen is felt — **stop palpating and refer** (friable, ruptures easily); describe cm below costal margin. Enlarged with mononucleosis, trauma, leukemia/lymphomas, portal hypertension, HIV; consider malaria with fever + travel.
- **Kidneys** [Jarvis p. 553]: "duck-bill" hands at flank, firm pressure, deep breath — right kidney occasionally felt as round smooth mass sliding between fingers (normal); left kidney sits 1 cm higher and is not normally palpable.
- **Aorta** [Jarvis p. 553]: opposing thumb and fingers, upper abdomen slightly left of midline; normally **2.5–4 cm** wide, pulsating anteriorly. Widened/lateral pulsation pushing fingers apart = abdominal aortic aneurysm; palpation accuracy is poor due to skin/adipose tissue and retroitoneal position.

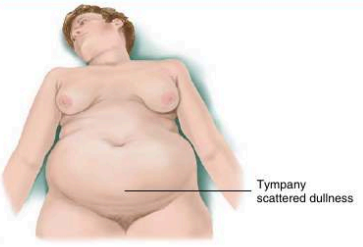
Special tests (Jarvis; her slides cover only rebound tenderness) [Jarvis pp. 555–556]:

- **Rebound tenderness**: site remote from painful area, hand 90° to abdomen, push down slowly and deeply, lift up quickly. Normal = no pain on release. Pain on release = peritoneal inflammation (e.g., appendicitis). Do this test **at the end** of the exam. Localized **cough tenderness** also signals peritoneal irritation. RLQ rebound with LLQ pressure = **Blumberg sign** (may indicate appendicitis; refer for CT).
- **Inspiratory arrest (Murphy sign)**: fingers under liver border, deep breath. Positive = sharp pain and abrupt stop of inspiration midway = cholecystitis. Less accurate in patients >60 y (25% have no abdominal tenderness).
- **McBurney point tenderness**: point 1.5–2 inches from the anterior superior iliac spine along the line to the umbilicus; maximal tenderness here with appendicitis.
- **Iliopsoas muscle test**: supine, lift right leg straight up flexing at hip, push down on distal thigh as person resists. Positive = RLQ pain (inflamed/perforated appendix).
- **Obturator test**: right leg flexed 90° at hip and knee, rotate internally/externally. Positive = pain (inflamed appendix irritating obturator muscle); less specific.
- **Alvarado (MANTRELS) score**: Migration to right iliac fossa 1, Anorexia 1, Nausea/vomiting 1, Tenderness RLQ 2, Rebound 1, Elevation of temperature (oral ≥37.3°C) 1, Leukocytosis (WBC >10,000/μL) 2, Shift to the left (>75% neutrophils) 1 — total 10. Score ≤4 significantly decreases probability of appendicitis; ≥7 increases it.

7. Abnormal abdominal findings

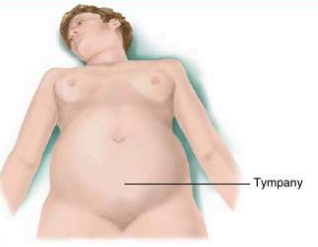
ABNORMAL FINDINGS

TABLE 22.1 Abdominal Distention



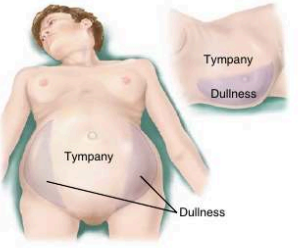
Obesity

Inspection. Uniformly rounded. Umbilicus sunken (it adheres to peritoneum, layers of fat are superficial to it).
Auscultation. Normal bowel sounds.
Percussion. Tympany. Scattered dullness over adipose tissue.
Palpation. Normal. May be hard to feel through thick abdominal wall.



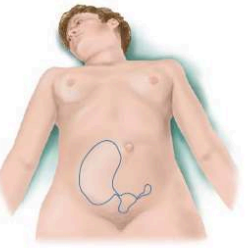
Air or Gas

Inspection. Single round curve.
Auscultation. Depends on cause of gas (e.g., decreased or absent bowel sounds with ileus); hyperactive with early intestinal obstruction.
Percussion. Tympany over large area.
Palpation. May have muscle spasm of abdominal wall.



Ascites

Inspection. Single curve. Everted umbilicus. Bulging flanks when supine. Taut, glistening skin due to recent weight gain; increase in abdominal girth.
Auscultation. Normal bowel sounds over intestines. Diminished over ascitic fluid.
Percussion. Tympany at top where intestines float. Dull over fluid. Produces fluid wave and shifting dullness.
Palpation. Taut skin and increased intra-abdominal pressure limit palpation.



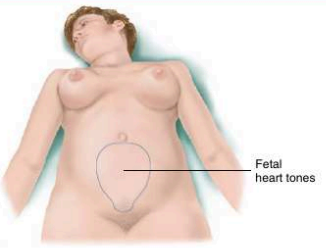
Ovarian Cyst (Large)

Inspection. Curve in lower half of abdomen, toward midline. Everted umbilicus.
Auscultation. Normal bowel sounds over upper abdomen where intestines pushed superiorly.
Percussion. Top dull over fluid. Intestines pushed superiorly. Large cyst produces fluid wave and shifting dullness.
Palpation. Transmits aortic pulsation, whereas ascites does not.

Continued

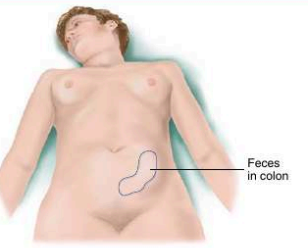
Table 22.1 Abdominal distention [Jarvis p. 563]

TABLE 22.1 Abdominal Distention—cont'd



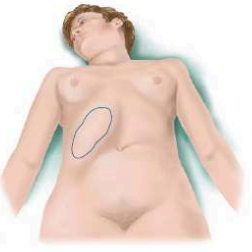
Pregnancy[a]

Inspection. Single curve. Umbilicus protruding. Breasts engorged.
Auscultation. Fetal heart tones. Bowel sounds diminished.
Percussion. Tympany over intestines. Dull over enlarging uterus.
Palpation. Uterine fundus. Fetal parts. Fetal movements.



Feces

Inspection. Localized distention.
Auscultation. Normal bowel sounds.
Percussion. Tympany predominates. Scattered dullness over fecal mass.
Palpation. Plastic-like or ropelike mass with feces in intestines.



◀ Tumor

Inspection. Localized distention.
Auscultation. Normal bowel sounds.
Percussion. Dull over mass if reaches up to skin surface.
Palpation. Define borders. Distinguish from enlarged organ or normally palpable structure.

[a]Obviously a normal finding, pregnancy is included for comparison of conditions causing abdominal distention.

Table 22.1 cont'd [Jarvis p. 564]

TABLE 22.2 Clinical Portrait of Intestinal Obstruction

Patient History and Symptoms

- History of previous abdominal surgery with adhesions
- Vomiting
- Fever
- Absence of stool or gas passage
- Colicky pain from strong peristalsis above the obstrucion

Physical Examination Findings

- Restless, ill-appearing patient
- Distended abdomen
- Hyperactive bowel sounds in early obstruction, hypoactive or silent in late obstruction
- Tenderness to palpation
- Hypovolemic shock due to dehydration and sepsis may occur (increased pulse, decrease BP, cool skin, dry mucous membranes)

Diagnostic Tests

- Laboratory: Evidence of dehydration, loss of electrolytes, and possibly sepsis
- Radiology: Accumulation of fluid and gas in bowel proximal to (above) obstruction.[5]



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Table 22.2 [Jarvis p. 565]

Abnormal findings [Slide 27]: abdominal distention; abdominal pain; hernia; abnormal bowel sounds; enlarged organs.

Jarvis Table 22.1 — Abdominal Distention [Jarvis pp. 563–564]:

Cause	Inspection	Percussion / palpation
Obesity	Uniformly rounded; umbilicus sunken	Tympany with scattered dullness over fat; palpation normal/hard through wall
Air or gas	Single round curve	Tympany over large area; muscle spasm possible
Ascites	Single curve, everted umbilicus, bulging flanks, taut glistening skin, increased girth	Tympany on top, dull over fluid; fluid wave + shifting dullness; taut wall limits palpation
Large ovarian cyst	Curve in lower half toward midline, everted umbilicus	Dull over fluid; fluid wave + shifting dullness; **transmits aortic pulsation (ascites does not)**

Table 22.1 cont'd [Jarvis p. 564]:

Cause	Inspection	Percussion / palpation
Pregnancy (comparison)	Single curve, protruding umbilicus, engorged breasts	Fetal heart tones; dull over uterus; uterine fundus, fetal parts/movements
Feces	Localized distention	Tympany predominates, scattered dullness; plastic/ropelike mass
Tumor	Localized distention	Dull over mass if reaches skin surface; define borders vs. organ

Abnormal bowel sounds & intestinal obstruction [Slide 17; Jarvis Table 22.2, p. 565]:

- Hyperactive (loud, high-pitched, rushing, tinkling) = increased motility: early obstruction, gastroenteritis, diarrhea, laxatives, subsiding paralytic ileus.
- Hypoactive/absent: abdominal surgery, peritonitis, paralytic ileus, complete obstruction.
- **Clinical portrait of intestinal obstruction** [Jarvis Table 22.2]: history of prior abdominal surgery with adhesions; vomiting; fever; absence of stool or gas; colicky pain. Exam: restless, ill-appearing; distended abdomen; **hyperactive sounds early → hypoactive/silent late**; tenderness; possible hypovolemic shock (increased pulse, decreased BP, cool skin, dry mucous membranes). Tests: dehydration/electrolyte loss/sepsis labs; radiology shows fluid and gas proximal to obstruction.

Other abnormal findings named in Jarvis [Jarvis pp. 543–548, 551, 553]:

- **Hernia** = protrusion of abdominal viscus through abnormal opening in muscle wall [Jarvis p. 543].
- Enlarged organs: liver >1–2 cm below right costal margin (firm may indicate cirrhosis); palpable spleen (3× normal size to be felt); widened aorta (aneurysm).
- Rigidity (involuntary, boardlike) = acute peritoneal inflammation; large masses; tenderness.

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8. Developmental considerations and sample charting (abdomen)

Jarvis Developmental Competence [Jarvis pp. 557–559]:

Infant [Jarvis pp. 557–558]:

- Contour protuberant (immature musculature); scaphoid = dehydration. Fine superficial venous pattern normal in lightly pigmented children to puberty.
- Umbilical cord: at birth white, **two arteries and one vein** in Wharton's jelly; stump dries in a week, falls off by 10–14 days; skin covers by 3–4 weeks. One artery only = risk for congenital defects.
- Umbilical hernia appears at 2–3 weeks, most prominent with crying, max at 1 month (up to 2.5 cm/1 in), usually gone by 1 year. Refer if >2.5 cm; asymptomatic hernias monitored to 4–5 years. **Diastasis recti** = midline bulge from separated rectus muscles; more common with Black infants; refer if lasting >6 years.
- Auscultation: bowel sounds only (metallic tinkling); no vascular sounds. Percussion: tympany over stomach, dullness over liver and bladder (may extend to umbilicus).
- Palpation: flex knees; pacifier if crying; abdomen soft and supple; liver edge at or 1–2 cm below right costal margin is normal; spleen tip, both kidneys, bladder, cecum, and sigmoid (sausage-like, left inguinal area) may be palpable.
- First stool = greenish-black **meconium** within 24 h; by day 4 breastfed = golden yellow, pasty, sour-milk smell; formula-fed = brown-yellow, firmer, fecal smelling.

Child [Jarvis p. 558]:

- <4 y protuberant supine and standing; after 4 y potbelly only when standing (lumbar lordosis), flat supine; scaphoid = dehydration/malnutrition.
- Respirations remain abdominal until **7 years** of age; absence of abdominal respirations <7 y = peritoneal inflammation.
- Palpate with child on parent's lap, knees flexed, head slightly up, "pant like a dog"; rest flat palm first. Liver edge 1–2 cm below costal margin normal; spleen edge and kidney tips commonly felt.
- Young children often answer "yes" to pain — use objective signs: change in cry pitch, grimacing, moving away, guarding.
- School-age: slim abdomen; adolescents — sensitivity to exposure, use adequate draping or keep in own clothing; findings same as adult.

Aging adult [Jarvis p. 559]:

- Increased subcutaneous fat on abdomen/hips (redistribution away from extremities); thinner, less toned musculature — peristalsis may be visible without obesity.
- Organs easier to palpate through the thinner wall: liver edge at or just below costal margin (1–2 cm below with inhalation with depressed diaphragm); kidneys easier to palpate.
- Abdominal rigidity with acute abdomen is **less common** in aging; they often complain of **less pain** than younger patients.

Sample charting (verbatim objective wording from Jarvis case) [Jarvis p. 561, Clinical Case Study 2]: "Abdomen is distended with everted umbilicus and bulging flanks… Prominent dilated venous pattern present… Bowel sounds present. No vascular sounds… tympany present at dome of abdomen, dullness over flanks. Shifting dullness present. Positive fluid wave present… Liver feels firm, nodular, nontender." — this models the terminology your sheet requires.

⚠ **Note:** Your Module 4 Documentation Sheet is a graded assignment — it is not filled in here. It keys its abdomen section to Jarvis: Inspection p. 542 (contour, symmetry, umbilicus, pulsations, skin), Auscultation p. 545 (bowel sounds: present/hyperactive/hypoactive), Percussion p. 546 (overall sound), Light Palpation p. 548 (smooth, soft, rigid, guarding, tenderness). Use the terminology in Sections 3–6 above for the "detail those marked abnormal" lines.

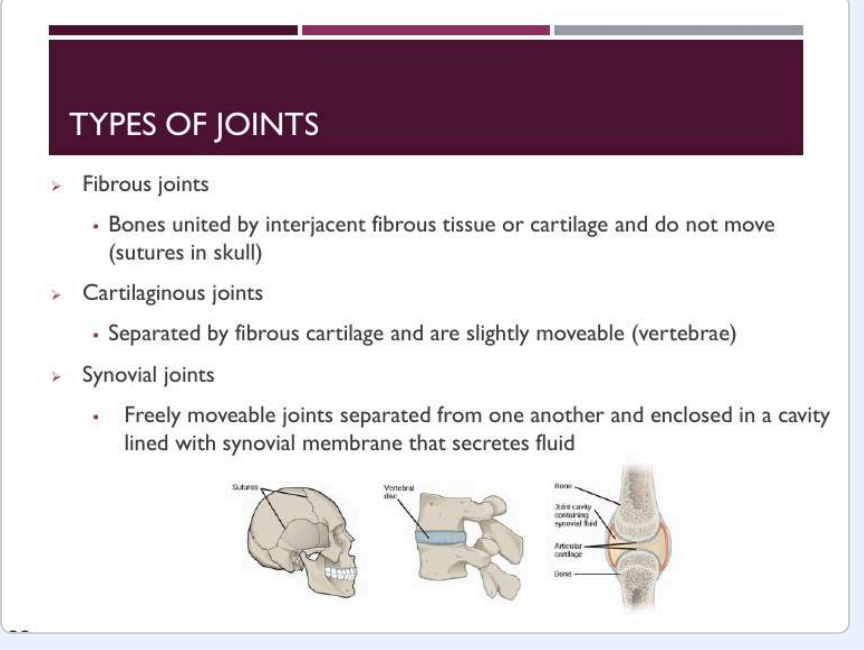
9. Musculoskeletal: overview and types of joints

Slide 28 — Musculoskeletal System

- Assessment of joints, muscles, and bones
- Functional assessment of ADLs [Slide 28]

Slide 29 — Types of Joints

- **Fibrous:** bones united by interjacent fibrous tissue or cartilage; do not move (sutures in skull)
- **Cartilaginous:** separated by fibrous cartilage; slightly moveable (vertebrae)
- **Synovial:** freely moveable; bones separated and enclosed in a cavity lined with synovial membrane that secretes fluid [Slide 29]



Types of joints [Slide 29]

Jarvis fill-in — purpose and approach. Purposes of the musculoskeletal exam are to assess function for ADLs and to screen for abnormalities. Much ADL data already comes from the history; add observations during the exam: gait, posture, how the person sits/rises from a chair, takes off a jacket, manipulates a small object such as a pen, and rises from supine [Jarvis p. 585].

- **Screening exam** suffices for most people: inspection and palpation of joints integrated with each body region, observation of ROM as the person moves, and age-specific screening (e.g., Ortolani sign for infants, scoliosis screening for adolescents) [Jarvis p. 585]
- **Complete exam** (as in this chapter) for articular disease, MSK symptoms, or ADL problems [Jarvis p. 585]
- Make the person comfortable; drape to visualize the part; go head to toe, proximal to distal [Jarvis p. 585]
- Support each joint at rest so muscles are soft and relaxed; use firm support and gentle movement around inflamed areas [Jarvis p. 585]
- Compare corresponding paired joints — expect symmetry of structure and function [Jarvis p. 585]
- **Equipment:** tape measure, skin marking pen, goniometer (occasional) [Jarvis p. 585]

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10. Inspection and palpation of joints

Slide 30 — Inspection

- Note size and contour of joints
- Inspect skin and tissue overlying joints for color, swelling, masses, or deformity [Slide 30]

Slide 31 — Palpation

- Palpate each joint for: skin temperature, muscles, bony articulations, joint capsule
- Joints normally non-tender when palpated [Slide 31]

Jarvis fill-in — inspection. Note the size and contour of every joint. Inspect skin and tissues over joints for color, swelling, and any masses or deformity. Swelling is significant and signals joint irritation; use the contralateral side for comparison [Jarvis p. 585].

Causes of swelling [Jarvis pp. 585–586]: excess joint fluid (effusion), thickening of the synovial lining, inflammation of surrounding soft tissue (bursae, tendons), or bony enlargement.

Deformities [Jarvis p. 586]: fracture (break in a bone); dislocation (complete loss of contact between the two bones); subluxation (bones stay in contact but alignment is off); contracture (muscle shortening → limited ROM); ankylosis (stiffness or fixation of a joint).

Jarvis fill-in — palpation. Palpate each joint, including its skin for temperature, its muscles, bony articulations, and area of joint capsule. Notice heat, tenderness, swelling, or masses. Joints normally are not tender; if tenderness occurs, localize it (skin, muscles, bursae, ligaments, tendons, fat pads, joint capsule) [Jarvis p. 586].

- The synovial membrane normally is not palpable; when thickened it feels "doughy" or "boggy" [Jarvis p. 586]
- A small amount of fluid is present in a normal joint but is not palpable; palpable fluid is abnormal and shifts to cause visible bulging on another side of the sac [Jarvis p. 586]
- Warmth and tenderness signal inflammation [Jarvis p. 586]

11. Range of motion

Slide 32 — Range of Motion

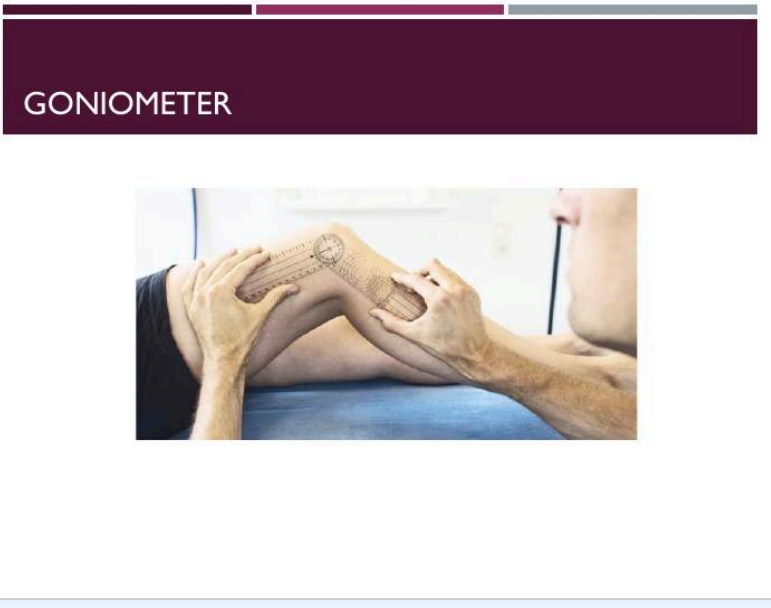
- Have client move through active ROM
- If limitation occurs, move joint through passive ROM
- Compare joints bilaterally
- Normally no tenderness, pain, or crepitation is present [Slide 32]

Slide 33 — Movements Produced by Skeletal Muscles

- Flexion: bending limb at joint
- Extension: straightening limb at joint
- Abduction: moving limb away from midline
- Adduction: moving limb toward midline
- Pronation: forearm turned so palm is down
- Supination: forearm turned so palm is up
- Circumduction: moving arm in a circle around the shoulder
- Inversion: sole of foot turned inward at ankle
- Eversion: sole of foot turned outward at ankle
- Rotation: moving head around central axis [Slide 33]

Slide 34 — Goniometer

Slide 34: picture of a goniometer measuring a knee [Slide 34].



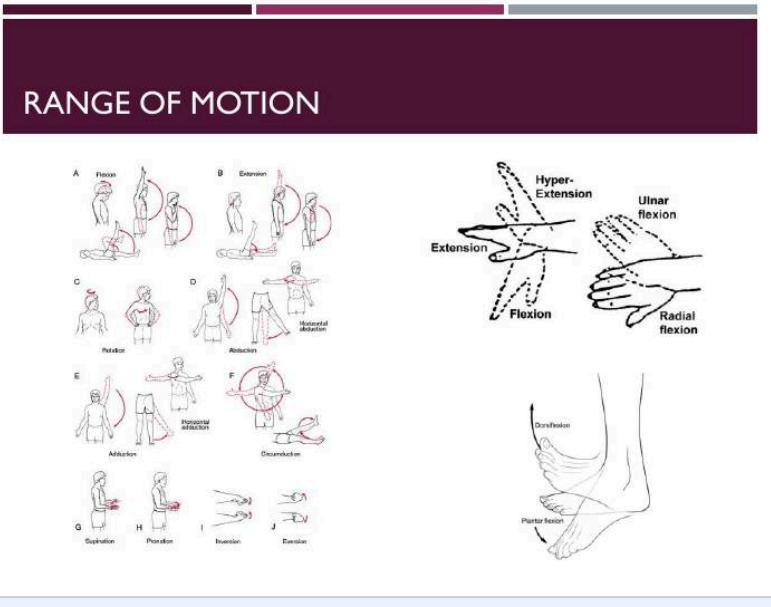
Goniometer [Slide 34]

Slide 35 — ROM Head-to-Toe Approach

TMJ • Cervical spine • Shoulder • Elbow • Wrist & Hand • Hip • Knee • Ankle & Foot • Spine [Slide 35]

Slide 36 — Range of Motion

Slide 36: title only / picture of ROM demonstration — no written points; filled from Jarvis below [Slide 36].



Range of motion [Slide 36]

Jarvis fill-in — technique. Ask for active (voluntary) ROM while modeling the movements yourself; stabilize the body area proximal to the part being moved. If you see a limitation, gently attempt passive motion with muscles relaxed, anchoring the joint with one hand while the other moves it to its limit. Normal ranges of active and passive motion should be the same [Jarvis p. 586].

Abnormal ROM. Limitation in ROM is the most sensitive sign of joint disease. Articular disease (inside the capsule, e.g., arthritis) causes swelling and tenderness around the whole joint and limits all planes of ROM in both active and passive motion. Extra-articular disease (specific tendon, ligament, nerve) causes swelling and tenderness at one spot and affects certain planes, especially active motion [Jarvis p. 586].

Crepitation is audible and palpable crunching or grating with movement, from roughened articular surfaces (e.g., RA). Do not confuse it with the normal discrete "crack" when a tendon or ligament slips over bone (e.g., during a knee bend) [Jarvis p. 586].

ROM Motion Terms

Term	Meaning
Flexion	Bending limb at joint
Extension	Straightening limb at joint
Hyperextension	Extension beyond straight (e.g., chin lifted toward ceiling)
Abduction	Moving limb away from midline
Adduction	Moving limb toward midline
Pronation	Forearm turned so palm is down
Supination	Forearm turned so palm is up
Circumduction	Moving arm in a circle around the shoulder
Inversion	Sole of foot turned inward at ankle
Eversion	Sole of foot turned outward at ankle
Rotation	Moving head/limb around a central axis

[Slide 33]; hyperextension/other ranges as measured per joint [Jarvis pp. 588–603]

ROM by Joint — Expected Ranges (Jarvis)

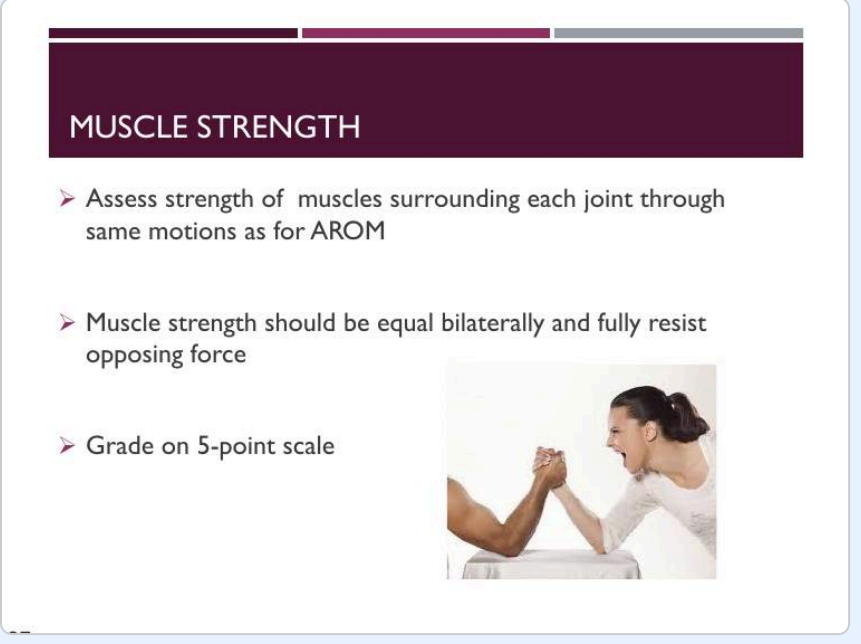
Joint	Motions tested	Expected range
TMJ	Open mouth; lateral jaw; protrude	Open 3–6 cm (three fingers sideways); lateral 1–2 cm; protrudes without deviation [Jarvis p. 587]
Cervical spine	Flexion / hyperextension / lateral bending / rotation	45° / 55° / 40° / 70° [Jarvis p. 588]
Shoulder	Flexion / hyperextension / abduction / adduction / int. rotation / ext. rotation	180° / up to 50° / 180° / 50° / 90° / 90° [Jarvis pp. 589–590]
Elbow	Flexion / extension; pronation / supination	150–160° / 0° (±5–10°); hand touches table both ways [Jarvis p. 591]
Wrist & hand	Wrist hyperext / palmar flexion; MCP flexion / hyperext; ulnar / radial deviation; abduction	70° / 90°; 90° / 30°; 50–60° / 20°; abduction 20°, fist tight [Jarvis pp. 593–594]
Hip	Flexion (knee straight) / flexion (knee bent) / int. rotation / ext. rotation / abduction / adduction / hyperext	90° / 120° / 40° / 45° / 40–45° / 20–30° / 15° [Jarvis p. 596]
Knee	Flexion / extension	130–150° / 0° (some hyperextend 15°) [Jarvis p. 599]
Ankle & foot	Plantar flexion / dorsiflexion / eversion / inversion; toes flex/extend	45° / 20° / 20° / 30° [Jarvis p. 601]
Spine	Flexion / hyperext / lateral bending / rotation	75–90° / 30° / 35° / 30° each side [Jarvis pp. 602–603]

> ⚠ **Note:** Jarvis states these are the expected ranges for each joint; her slides give only the motion terms and the head-to-toe order, not degrees [Slide 35]. Some healthy people lack 5–10° of elbow extension or have 5–10° hyperextension [Jarvis p. 591].

12. Muscle strength grading

Slide 37 — Muscle Strength

- Assess strength of muscles surrounding each joint through the same motions as for AROM
- Muscle strength should be equal bilaterally and fully resist opposing force
- Grade on 5-point scale [Slide 37]



Muscle strength [Slide 37]

Normal Range of Findings				Abnormal Findings
Grade	**Description**	**% Normal**	**Assessment**	
5	Full ROM against gravity, full resistance	100	Normal	
4	Full ROM against gravity, some resistance	75	Good	
3	Full ROM with gravity	50	Fair	
2	Full ROM with gravity eliminated (passive motion)	25	Poor	
1	Slight contraction	10	Trace	
0	No contraction	0	Zero	

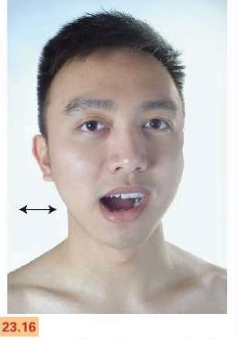
Temporomandibular Joint

With the person seated, **inspect** the area just anterior to the ear. Place the tips of your first two fingers in front of each ear and ask the person to open and close the mouth. Drop your fingers into the depressed area over the joint and note smooth motion of the mandible. An audible and palpable snap or click occurs in many healthy people as the mouth opens (Fig. 23.15). Then ask the person to:

Instructions to Person	Motion and Expected Range	Abnormal Findings
• Open mouth as wide as possible.	Vertical motion. You can measure the space between the upper and lower incisors. Normal is 3 to 6 cm or three fingers inserted sideways.	Swelling looks like a round bulge over the joint, although it must be moderate or marked to be visible. Crepitus and pain occur with TMJ dysfunction during movement or chewing. Malocclusion of teeth also causes palpable crepitus or an audible click.
• Partially open mouth, protrude lower jaw, and move it side to side.	Lateral motion. Normal extent is 1 to 2 cm (Fig. 23.16).	Decreased ROM occurs with TMJ inflammation and arthritis.
• Stick out lower jaw.	Protrude without deviation.	Lateral motion may be lost earlier and more significantly than vertical motion.



23.15



23.16

Palpate the contracted temporalis and masseter muscles as the person clenches the teeth. Compare right and left sides for size, firmness, and strength. Ask the person to move the jaw forward and laterally against your resistance and open mouth against resistance. This also tests the integrity of cranial nerve V (trigeminal).

TMJ dysfunction causes tenderness with palpation.

Objective Data

Muscle grading scale (0–5) and TMJ [Jarvis p. 587]

Jarvis fill-in. Test the strength of the prime-mover muscle groups for each joint. Repeat the motions elicited for active ROM; ask the person to flex and hold while you apply opposing force. Strength should be equal bilaterally and fully resist your opposing force. Muscle status and joint status are interdependent and should be interpreted together [Jarvis p. 586]. A wide variability of strength exists among people; use a grading system from no voluntary movement to full strength [Jarvis p. 586].

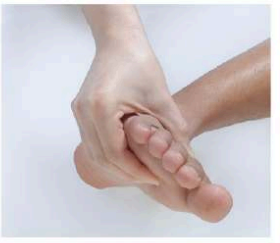
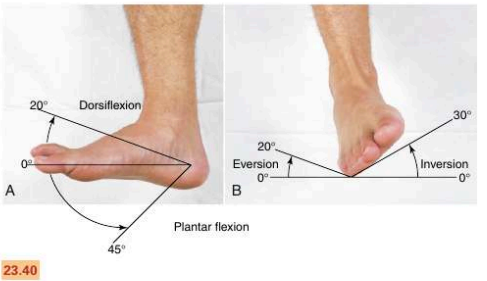
Muscle Strength Grading (Jarvis)

Grade	Description	% Normal	Assessment
5	Full ROM against gravity, full resistance	100	Normal
4	Full ROM against gravity, some resistance	75	Good
3	Full ROM with gravity	50	Fair
2	Full ROM with gravity eliminated (passive motion)	25	Poor
1	Slight contraction	10	Trace
0	No contraction	0	Zero

[Jarvis p. 587]

> ⚠ **Conflict:** Her slide says grade on a **5-point scale** [Slide 37]; Jarvis uses grades **0 through 5** (six levels, including 0 = no contraction) with % normal and descriptors [Jarvis p. 587]. Her documentation sheet directs you to place a **numeric value from 0–5** using the Muscle Grading Scale p. 587 — so use Jarvis's 0–5 numeric values on the assignment.

13. Spine and special tests

Normal Range of Findings	Abnormal Findings
 23.39	
Test ROM (Fig. 23.40) by asking the person to:	
Instructions to Person • Point toes toward the floor. • Point toes toward the nose. • Turn soles of feet out, then in. (Stabilize ankle with one hand and hold heel with the other to test the subtalar joint.) • Flex and straighten toes.	**Motion and Expected Range** Plantar flexion of 45 degrees. Dorsiflexion of 20 degrees (Fig. 23.40A). Eversion of 20 degrees. Inversion of 30 degrees (Fig. 23.40B).	Limited ROM. Pain with motion.
Assess **muscle strength** by asking the person to maintain dorsiflexion and plantar flexion against your resistance.		Unable to hold flexion.
 23.40		
Spine The person should be standing, draped in a gown open at the back. Place yourself far enough back so you can see the entire back. **Inspect** and note whether the spine is straight (1) by following an imaginary vertical line from the head through the spinous processes and down through the gluteal cleft; and (2) by noting equal horizontal positions for the shoulders, scapulae, iliac crests, and gluteal folds and equal spaces between the arm and lateral thorax on the two sides (Fig. 23.41A). The person's knees and feet should be aligned with the trunk and pointing forward. From the side note the normal convex thoracic curve and concave lumbar curve (Fig. 23.41B). An enhanced thoracic curve, or kyphosis, is common in aging people. A pronounced lumbar curve, or lordosis, is common in obese people.		A difference in shoulder elevation and in level of scapulae and iliac crests occurs with scoliosis (see Table 23.7, Spine Abnormalities, p. 624). Lateral tilting and forward bending occur with a herniated nucleus pulposus (see Table 23.7).

Objective Data

Fig. 23.39–23.40: spine inspection and ROM [Jarvis p. 601]

> No written slide points for this section appear in slides 28–37; the spine is listed only as a ROM region on the head-to-toe approach [Slide 35]. Content below is from Jarvis.

Spine inspection. Person standing, gown open at back. Spine is straight: an imaginary vertical line runs from the head through the spinous processes and gluteal cleft; shoulders, scapulae, iliac crests, and gluteal folds are level; spaces between arms and lateral thorax are equal. From the side: normal convex thoracic and concave lumbar curves [Jarvis p. 601].

- **Kyphosis:** enhanced thoracic curve — common in aging people; also common in adolescence with chronic poor posture [Jarvis pp. 601, 609]
- **Lordosis:** pronounced lumbar curve — common in obese people; common in childhood with protuberant abdomen; marked with muscular dystrophy and rickets [Jarvis pp. 601, 608]
- **Scoliosis:** difference in shoulder elevation and levels of scapulae and iliac crests; rib hump on one side with forward bend [Jarvis pp. 601, 610]

Palpation. Spinous processes straight and nontender; paravertebral muscles firm, no tenderness or spasm [Jarvis p. 602].

Spine ROM: flexion 75–90° (lumbar concavity disappears, back forms a single convex C), hyperextension 30°, lateral bending 35°, rotation 30° bilaterally. Marking dots on spinous processes while bending should form a straight vertical line on standing; a slight S-shape indicates spinal curve [Jarvis pp. 602–603]. Finally, walk on toes, then on heels [Jarvis p. 603].

Special Tests

Test	How / what it checks	Positive / abnormal
Phalen	Hold both hands back to back, wrists flexed 90° for 60 seconds [Jarvis p. 595]	Reproduces numbness and burning = carpal tunnel syndrome
Tinel sign	Direct percussion over median nerve at the wrist [Jarvis p. 595]	Burning/tingling along nerve distribution = positive, carpal tunnel
Bulge sign	Stroke medial knee 2–3×, tap lateral side, watch medial hollow [Jarvis pp. 597–598]	Visible fluid wave = effusion of 4–8 mL; high risk for knee pain/progressive OA
Ballottement of patella	Compress suprapatellar pouch, push patella sharply against femur [Jarvis p. 598]	Tap heard as patella bumps femoral condyles = fluid present (larger amounts)

Test	How / what it checks	Positive / abnormal
McMurray	Supine; flex knee and hip, rotate leg, apply valgus stress, slowly extend [Jarvis p. 599]	"Click" felt/heard = torn meniscus; refer to orthopedics
Lasègue / straight leg raising	Raise extended leg just short of pain, then dorsiflex foot; then raise unaffected leg [Jarvis p. 604]	Reproduces/worsens sciatic pain, worse with dorsiflexion = herniated disc
Thomas test	Flex one knee to chest with other leg straight, supine [Jarvis p. 596]	Opposite thigh lifts off table = hip flexion deformity
Ortolani / Allis	Infant hips: adduct then abduct/lift knees (Ortolani); compare knee heights with knees flexed (Allis) [Jarvis pp. 605–606]	Clunk as femoral head reseats = positive Ortolani; one knee lower = positive Allis

> ⚠ **Note:** The **Barlow maneuver** is not described in the assigned Jarvis pages (pp. 585–613); Jarvis describes the **Ortolani maneuver** (done at every professional visit until age 1 year) and the **Allis test** for hip dislocation [Jarvis pp. 605–606].

> ⚠ **Note:** Do not attempt cervical motions if you suspect neck trauma [Jarvis p. 588 footnote].

Other measures: Leg length discrepancy — true length from anterior iliac spine to medial malleolus; normally equal or within 1 cm. Apparent length from umbilicus to medial malleolus; unequal apparent lengths with equal true lengths occur with pelvic obliquity or hip adduction/flexion deformity [Jarvis p. 604]. Ottawa knee/ankle rules govern imaging referral [Jarvis pp. 600, 603].

14. Developmental considerations and sample charting (MSK)

Normal Range of Findings	Abnormal Findings
The U.S. Preventive Services Task Force[29] does not support the routine screening of asymptomatic adolescents for idiopathic scoliosis. They found that most cases detected through screening did not progress to clinically significant scoliosis. In addition, the harm of unnecessary imaging, follow-up visits, and psychological adverse effects of brace wear did not offset any potential benefits of screening. However, when idiopathic scoliosis is discovered incidentally or when the teen or parent expresses concern, clinicians should be prepared to evaluate. **Screen for scoliosis** with the *forward bend test*. Seat yourself behind the child so that you can easily view the spine. Ask the adolescent to stand with feet shoulder width apart and bend forward slowly to touch the toes. Expect a straight vertical spine while standing and also while bending forward. Posterior ribs should be symmetric, with equal elevation of shoulders, scapulae, and iliac crests.	Scoliosis is most apparent during the preadolescent growth spurt; marked asymmetry suggests scoliosis—ribs hump up on one side as child bends forward, with unequal landmark elevation (see Table 23.7).

Objective Data

Pregnancy

Proceed through the examination described in the adult section. Expected postural changes in pregnancy include progressive lordosis and, toward the third trimester, anterior cervical flexion, kyphosis, and slumped shoulders (Fig. 23.53A). When the pregnancy is at term, the protuberant abdomen and relaxed mobility in the joints create the characteristic "waddling" gait (Fig. 23.53B).



23.53

The Aging Adult

Postural changes include a decrease in height, more apparent in the 70s and 80s (Fig. 23.54). The expression "lengthening of the arm-trunk axis" describes this shortening of the trunk with comparatively long extremities. Kyphosis is

The aging adult: posture changes [Jarvis p. 610]

> No written slide points for this section appear in slides 28–37; filled from Jarvis below.

General. Keep a chart of the usual sequence of motor development; a questionnaire such as the Ages and Stages Questionnaires screens fine and gross motor skills. Muscle tone, resting posture, and motor activity overlap with the neurologic exam (Chapter 24) [Jarvis p. 605].

- **Infants:** examine undressed on a warming table. Feet/legs — note flexible vs fixed positional varus/valgus; metatarsus adductus usually resolves by age 3 years; check tibial torsion. Hips — Ortolani (every visit until 1 year), Allis, gluteal folds. Hands/arms — polydactyly, syndactyly, single palmar crease; palpate clavicles (most common birth fracture); equal arm ROM with Moro reflex. Back — normal single C-curve at birth; concave cervical curve by 2 months when infant lifts head prone; tuft of hair/dimple/mass may indicate spina bifida or dermoid sinus. Muscle strength — lift under axillae; normal infant wedges securely [Jarvis pp. 605–607]
- **Preschool/school-age:** observe during spontaneous play. Shoulders level within 1 cm, scapulae symmetric; lordosis common. Genu varum normal for 1 year after walking begins (>2.5 cm between knees with malleoli together); genu valgum normal at 2–3½ years (>2.5 cm between malleoli with knees together). Pes planus normal in infants, resolves by 3 years; pronation common 12–30 months; pronation/lack of arch after age 3 is abnormal. In-toeing often forefoot adduction, self-correcting if flexible. Gait: broad-based at 1–2 years, base narrows from 3 years. Watch for radial head subluxation (ages 2–4) [Jarvis pp. 608–609]
- **Adolescents:** adult exam with special note to posture; kyphosis common with poor posture; sports injuries at peak. USPSTF does not support routine screening of asymptomatic adolescents for idiopathic scoliosis; when concerned, use the forward bend test — rib hump/unequal landmark elevation suggests scoliosis [Jarvis pp. 609–610]
- **Pregnancy:** progressive lordosis; third-trimester anterior cervical flexion, kyphosis, slumped shoulders; "waddling" gait at term [Jarvis p. 610]
- **Aging adult:** decreased height (apparent in 70s–80s), "lengthening of the arm-trunk axis," kyphosis with backward head tilt (figure 3 outline from the left), slight hip/knee flexion, peripheral fat loss with abdominal/hip deposition, prominent bony prominences. ROM and strength much the same as younger adults absent disease. **Get Up and Go Test:** rise from armchair, walk 10 feet, turn, return, sit — a healthy adult over 60 does this in under 12 seconds; 12 seconds or more = increased fall risk. Functional ADL assessment for advanced aging/arthritic/disability changes [Jarvis pp. 610–611]

Sample Charting [Jarvis p. 613]

SUBJECTIVE: States no joint pain, stiffness, swelling, or limitation. No muscle pain or weakness. No history of bone trauma or deformity. Able to manage all usual daily activities with no physical limitations. Occupation involves no musculoskeletal risk factors. Exercise pattern is brisk walk 1 mile 5×/week.

OBJECTIVE: Joints and muscles symmetric; no swelling, masses, deformity; normal spinal curvature. No tenderness to palpation of joints; no heat, swelling, or masses. Full ROM; movement smooth, no crepitus, no tenderness. Muscle strength —able to maintain flexion against resistance and without tenderness.

ASSESSMENT: Muscles and joints—healthy and functional

⚠ **Assignment link:** Her Module 4 documentation sheet uses Jarvis page numbers for MSK: inspection p. 585, palpation p. 586, muscle grading scale 0–5 p. 587 — matching the Jarvis citations above (form itself is left blank).

16. Quick-review tables

Key numbers

Item	Number	Source
Light palpation depth	~1 cm	[Slide 23; Jarvis p. 548]
Deep palpation depth	5–8 cm (2–3 in)	[Slide 24; Jarvis p. 549]
Normal bowel sounds	5–30/min, irregular (don't count)	[Jarvis p. 545]
Declare bowel sounds absent	After 5 min of auscultation	[Slide 17; Jarvis p. 545]
Normal aortic width (palpation)	2.5–4 cm	[Jarvis p. 553]
Liver enlarged if edge >	1–2 cm below right costal margin	[Jarvis p. 551]
Spleen palpable only when enlarged	3× normal size	[Jarvis p. 552]
Striae length	1–6 cm	[Jarvis p. 543]
Shifting dullness misses fluid <	500–1100 mL	[Jarvis p. 555]
Normal bruit in healthy people	4%–20% (usually <40 yr, celiac)	[Jarvis p. 546]

ROM reference (Jarvis, for orientation)

Region	Motion	Expected range
Cervical	Flexion / hyperextension	45° / 55°
Cervical	Lateral bending / rotation	40° / 70°
Shoulder	Flexion / hyperextension	180° / up to 50°
Elbow	Flexion / pronation–supination	150–160° / 90° each
Wrist	Hyperextension / palmar flexion	70° / 90°
Hip	Flexion (leg up / knee to chest)	90° / 120°
Knee	Flexion / extension	130–150° / 0° (or 15° hyperext)
Ankle	Plantar / dorsiflexion	45° / 20°
Spine	Flexion / hyperextension	75–90° / 30°

Source: [Jarvis pp. 588–603]. Her slide only specifies that ROM is compared bilaterally and normally pain-free [Slide 32].

Look-alike pairs

Pair	First	Second
Tympany vs dullness	Tympany: high-pitched, musical = air in stomach/intestine	Dullness: medium intensity, duration, pitch = solid organs, masses, fluid, adipose, distended bladder
(vs hyperresonance)	Hyperresonance: very loud, low pitch, long duration = gaseous distention [Slide 20; Jarvis p. 546]	
Hyper- vs hypoactive BS	Hyperactive: loud, high-pitched, rushing, tinkling = increased motility (early obstruction, gastroenteritis, diarrhea, laxatives)	Hypoactive/absent: surgery, peritonitis, paralytic ileus, complete obstruction [Slide 17]
Guarding vs rigidity	Voluntary guarding: bilateral, relaxes with exhalation (cold, tense, ticklish)	Involuntary rigidity: constant boardlike hardness; peritoneal inflammation; may be unilateral [Jarvis p. 548]

Spinal curve	What it is	Association
Kyphosis	Enhanced thoracic (convex) curve	Common with aging; adolescents with poor posture [Jarvis pp. 601, 609]
Lordosis	Pronounced lumbar (concave) curve	Common with obesity; pregnancy; childhood [Jarvis pp. 601, 610]
Scoliosis	Lateral spinal curvature	Unequal shoulder/scapula/iliac crest levels; rib hump on forward bend [Jarvis pp. 601, 610]

Motion pair	One	Opposite
Flexion / extension	Bending limb at joint	Straightening limb at joint
Abduction / adduction	Away from midline	Toward midline
Pronation / supination	Forearm turns palm down	Palm up
Inversion / eversion	Sole of foot inward (ankle)	Sole outward (ankle)

All motion definitions: [Slide 33]. Circumduction = arm circles around shoulder; rotation = head around central axis [Slide 33].

AROM vs PROM	Active	Passive
Who moves	Client moves voluntarily (you model it)	You move the part with muscles relaxed
When	First, always	Only if AROM shows limitation
Normal	Ranges of active and passive motion should be the same [Slide 32; Jarvis p. 586]	

Muscle grading scale (doc sheet points here)

Grade	Description	Label
5	Full ROM against gravity, full resistance	Normal
4	Full ROM against gravity, some resistance	Good
3	Full ROM with gravity	Fair
2	Full ROM with gravity eliminated	Poor
1	Slight contraction	Trace
0	No contraction	Zero

[Jarvis p. 587]; required on her sheet as numeric 0–5 per joint group [doc sheet].

17. Likely test traps

- **Abdominal exam order is inspection → auscultation → percussion → palpation** — auscultate second because percussion/palpation can alter peristalsis [Slide 6; Slide 17; Jarvis p. 545].
- Start bowel sounds in the **RLQ** (ileocecal valve area) with the **diaphragm** [Slide 17; Jarvis p. 545].
- Bowel sounds are called **absent only after 5 minutes** of continuous auscultation — a common test item [Slide 17; Jarvis p. 545].
- Hyperactive sounds = **increased** motility (early obstruction, gastroenteritis, diarrhea, laxatives, subsiding paralytic ileus); hypoactive/absent = surgery, peritonitis, paralytic ileus, **complete** obstruction [Slide 17].
- Document bowel sounds as **Present / Hyperactive / Hypoactive** — that is the wording on her sheet [doc sheet; Slide 17].
- Auscultate for **bruits** over aorta, renal, iliac, femoral arteries — that exact list [Slide 19; Jarvis p. 546].
- **Percussion is NOT recommended** to determine liver span, splenomegaly, or bladder distention — but DO percuss the **costovertebral angle** for kidney tenderness [Slide 21; Jarvis p. 547].
- Normal percussion = **tympany predominates** (air rises supine); dullness over solid organs, masses, fluid, adipose, distended bladder; hyperresonance = gaseous distention [Slide 20; Jarvis p. 546].
- Light palpation ~**1 cm**, deep ~**5–8 cm**; bimanual technique for very large/obese abdomen [Slides 23–24; Jarvis pp. 548–549].
- Light palpation's purpose is **not** to find organs — it forms an overall impression of skin surface and superficial musculature [Slide 23].
- **Assess tender areas LAST** and watch nonverbal cues of discomfort [Slide 22; Jarvis p. 548].
- Rebound tenderness technique: hand at **90°**, press **slowly and deeply** in a non-tender area, withdraw **quickly**; normally **no pain** on release; pain = peritoneal irritation/inflammation [Slide 26; Jarvis p. 555].

- Normal contour = **flat to rounded**; scaphoid (caves in) and protuberant (distention) are abnormal — the doc sheet lists all four [Slide 11; Jarvis p. 542].
- Umbilicus normally **midline and inverted**; everted with ascites/umbilical hernia [Slide 12; Jarvis p. 543].
- **Striae** are a common pigment change (silvery-white, 1–6 cm) — often tested as "which skin finding is normal?" [Slide 13; Jarvis p. 543].
- Visible epigastric aortic pulsation can be **normal** in thin, relaxed people [Slide 15; Jarvis p. 544].
- MSK joint exam = **inspect then palpate** (temperature, muscles, bony articulations, joint capsule); joints are normally **non-tender** [Slides 30–31; Jarvis pp. 585–586].
- ROM: **active first**, passive only if limited, compare **bilaterally**; normally no tenderness, pain, or crepitation [Slide 32; Jarvis p. 586].
- Don't confuse **crepitation** (audible/palpal crunching = abnormal) with the normal discrete "crack" of a tendon slipping over bone [Jarvis p. 586; Slide 32].
- Muscle strength = equal bilaterally, fully resists opposing force; grade with the **0–5** scale (5 = full ROM against gravity + full resistance) [Slide 37; Jarvis p. 587; doc sheet].
- Her ROM chart is rated per joint group, Right vs Left, and muscle strength is a numeric value per muscle group — know which is ✔/X and which is numeric [doc sheet].

> ⚠ **Note:** Jarvis adds a caution not in her slides: if you feel an enlarged spleen, **stop palpating** — it is friable and can rupture [Jarvis p. 552]. Worth remembering for safety items.

18. Module 4 vocabulary (18 terms)

#	Term	Definition	Source
1	**aneurysm**	Abnormal widening/dilation of an artery; abdominal aortic aneurysm shows a widened aortic pulsation (>2.5–4 cm) with prominent lateral pulsation, and may produce a systolic bruit.	[Jarvis pp. 546, 553]
2	**ascites**	Free fluid in the peritoneal cavity; shows everted umbilicus, bulging flanks, taut glistening skin, tympany over the top of the abdomen with dullness over fluid, positive fluid wave and shifting dullness.	[Jarvis pp. 554, 563]
3	**borborygmi**	Loud hyperperistalsis — the "stomach growling" heard as hyperactive bowel sounds.	[Jarvis p. 545]
4	**bowel sounds**	Sounds from movement of air and fluid within the stomach and large/small intestine; normally high-pitched, gurgling, cascading, irregular, 5–30/min; judged present, hyperactive, or hypoactive.	[Slide 17; Jarvis p. 545]
5	**bruit**	Pulsatile blowing vascular sound (usually systolic, medium-to-low pitch) heard over an artery; occurs with stenosis, partial occlusion, or aneurysm.	[Slide 19; Jarvis p. 546]
6	**epigastrium**	Upper central region of the abdomen (nine-region scheme) just below the xiphoid; the area where aortic pulsations may normally be seen in thin, relaxed people.	[Slides 8, 15; Jarvis p. 544]
7	**hepatomegaly**	Abnormal enlargement of the liver — liver edge palpated more than 1–2 cm below the right costal margin (except with a depressed diaphragm).	[Jarvis p. 551]

#	Term	Definition	Source
8	**hernia**	Protrusion of abdominal viscera through an abnormal opening in the muscle wall; a listed abnormal abdominal finding.	[Slide 27; Jarvis p. 543]
9	**rebound tenderness**	Pain on quick release after slow, deep pressure at 90° in a non-tender area; confirms peritoneal irritation/inflammation (e.g., appendicitis); normal = no pain on release.	[Slide 26; Jarvis p. 555]
10	**splenomegaly**	Enlargement of the spleen; the spleen is normally not palpable and must be ~3× normal size to be felt; best detected by palpation, not percussion.	[Slide 21; Jarvis pp. 547, 552]
11	**striae**	Silvery-white, linear, jagged skin marks 1–6 cm long from broken elastic fibers after stretching (pregnancy, weight gain); pink/blue when recent; a common abdominal pigment change.	[Slide 13; Jarvis p. 543]
12	**tympany**	High-pitched, musical percussion note produced by air in the stomach and intestines; normally predominates over the abdomen.	[Slide 20; Jarvis p. 546]
13	**crepitation**	Audible and palpable crunching or grating accompanying joint movement when articular surfaces are roughened (e.g., RA); abnormal — not the normal discrete "crack" of tendon over bone.	[Slide 32; Jarvis p. 586]
14	**kyphosis**	Enhanced outward (thoracic) curve of the spine; common with aging and adolescent poor posture.	[Jarvis pp. 601, 609]
15	**lordosis**	Pronounced inward (lumbar) curve of the spine; common with obesity, pregnancy, and childhood.	[Jarvis pp. 601, 610]
16	**myalgias**	Muscle pain/aching — asked about in the MSK history ("no muscle pain or weakness" is a normal response).	[Jarvis p. 613]

#	Term	Definition	Source
17	**range of motion**	The arc of movement of a joint, tested first as active (voluntary) ROM and then passive if limited; normally no tenderness, pain, or crepitation and equal bilaterally.	[Slide 32; Jarvis p. 586]
18	**scoliosis**	Lateral curvature of the spine; seen as unequal shoulder, scapula, or iliac crest levels and a rib hump on the forward-bend test.	[Jarvis pp. 601, 610]